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Elucidating response effects of anammox-based nitrogen removal processes for municipal wastewater using big data analysis and automated machine learning

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Applications of anaerobic ammonium oxidation (anammox) processes for treating municipal wastewater have promising benefits in reducing carbon emissions and have been heavily investigated recently. However, the practical operation of mainstream anammox processes still suffers from unclear external regulation strategies. Therefore, this study utilized the H₂O automated machine learning (AutoML) algorithm and interpretable analysis to capture the internal relationships existing in the big dataset collected from anammox-based studies for treating municipal wastewater. The eXtreme Gradient Boosting and gradient boosting machine models automatically generated by the AutoML algorithm provided the most accurate prediction results ($R^2 = 0.814\text{--}0.993$). Moreover, the optimal models showed good generalization ability ($R^2 = 0.725\text{--}0.945$) for unseen data collected from this study. The appropriate ranges of operation conditions and influent characteristics response to the high removal efficiency of nitrogen pollutants and high nitrogen removal rate through the anammox reaction pathway were revealed by the one-dimensional, and two-dimensional partial dependence plots. This work would advance the understanding of how to improve the practical operation of mainstream anammox-based nitrogen removal processes for treating municipal wastewater.

Keywords Mainstream anammox, Automated machine learning, Interpretable analysis, Nitrogen removal efficiency, Meta-analysis

The problem of water eutrophication caused by excessive nitrogen and phosphorus emissions has been a global concern for a long time^{1,2}. The traditional activated sludge is the most commonly used biological process for removing nitrogen and phosphorus. However, eliminating nitrogen through the nitrification and denitrification process by the conventional activated sludge has been identified as an energy-intensive process³. Therefore, anaerobic ammonium oxidation (anammox), a new technology with low energy consumption, low sludge productivity, and a high nitrogen removal rate, has drawn increasing attention in recent years^{2,4–6}.

Unlike traditional two-step nitrification and denitrification processes, the anammox process uses nitrite and ammonia as substrates and directly oxidizes them to N₂⁷. Anammox is one of the most promising processes for saving the operating costs of wastewater treatment plants (WWTPs), reducing organic carbon requirements by 100%, aeration energy consumption by approximately 60%, and sludge production by approximately 90%⁸. However, the stable supply of nitrite to anammox remains an urgent problem. To provide nitrite steadily, several types of combined processes including simultaneous nitrification, anammox and denitrification^{9,10}, partial nitrification coupling with anammox (PNA)¹¹, partial denitrification coupling with anammox (PDA)^{12,13}, and more complex processes combining PNA and PDA processes^{14,15} have been investigated extremely. Moreover, these anammox processes for mainstream application face many technical bottlenecks such as low nitrogen concentration of municipal wastewater, high organic matter concentration¹⁶, and low water temperature¹⁷ in winter. To solve these issues, numerous attempts have been made to mainstream anammox processes from multiple operational factors such as influent characteristics, inoculation mode, temperature, reactor type, and hydraulic retention time. A large amount of data has been generated during these experiments. However, the

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Các thuật toán học máy ngày càng thu hút nhiều sự chú ý nhờ khả năng nhận diện các mối quan hệ nội tại trong những tập dữ liệu lớn và xây dựng các mô hình dự đoán tương ứng^{18,19}. Phân tích có thể giải thích và các phương pháp diễn giải đã được phát triển nhằm giải thích sâu hơn về các mô hình học máy đã huấn luyện^{20–22}. Chẳng hạn, Xu et al. (2022b)²³ đã phân tích các hiệu ứng đáp ứng của quá trình anammox đối với các loại kháng sinh khác nhau thông qua một mô hình học máy phổ quát (R² inline-eq-IEq10.9). Tương tự, Yang et al. (2023)²⁴ tập trung vào áp lực kim loại nặng lên quá trình anammox dựa trên các mô hình học máy. Hơn nữa, Liu et al. (2023a)²⁵ đã khảo sát sự phát thải N₂O từ các quy trình dựa trên quá trình anammox. Họ đã làm rõ mối quan hệ giữa phát thải N₂O và các yếu tố vận hành cũng như quần xã vi sinh vật thông qua các mô hình học máy và phân tích diễn giải tương ứng. Những nghiên cứu này đã chứng minh rằng các mô hình học máy thể hiện hiệu quả tốt trong phân tích dữ liệu lớn cho các quy trình dựa trên quá trình anammox^{23–25}.

Do đó, trong nghiên cứu này, thuật toán học máy tự động (AutoML) đã được áp dụng để dự đoán hiệu suất xử lý nitơ và diễn giải các hiệu ứng đáp ứng của các quy trình dựa trên quá trình anammox đối với nhiều yếu tố vận hành. Dữ liệu từ các nghiên cứu liên quan dựa trên quá trình anammox trước hết được phân tích để lựa chọn các biến đầu vào và đầu ra tiềm năng, tạo thành tập dữ liệu tích hợp để chạy thuật toán H₂O AutoML. Sau đó, các mô hình ứng viên do thuật toán H₂O AutoML tạo ra cho 7 biến đầu ra đã được so sánh. Ngoài ra, các thí nghiệm quy mô phòng thí nghiệm dựa trên quá trình anammox đã được tiến hành để kiểm tra hiệu suất dự đoán của các mô hình ứng viên. Dựa trên các mô hình tối ưu, phân tích diễn giải đã được thực hiện nhằm làm rõ hiệu ứng đáp ứng của các biến đầu ra đối với các biến đầu vào quan trọng. Những kết quả này làm nổi bật các mối quan hệ quan trọng nhưng từng bị bỏ qua ảnh hưởng đến hiệu quả xử lý dựa trên quá trình anammox, mang lại một bước tiến đáng kể hướng tới việc cải tiến các quy trình dòng chính dựa trên quá trình anammox.

Vật liệu và phương pháp

internal knowledge about anammox-based biological nitrogen removal processes remains to be explored from these big data.

Machine learning algorithms have drawn increasing attention due to their ability to identify the internal relationships in big datasets and create corresponding prediction models^{18,19}. Explainable analysis and interpretable methods have been developed to further explain the trained machine learning models^{20–22}. For instance, Xu et al. (2022b)²³ analyzed response effects of anammox processes to different antibiotics through a universal machine learning model ($R^2 > 0.9$). Similarly, Yang et al. (2023)²⁴ focused on the heavy metal stress on anammox processes based on machine learning models. Moreover, Liu et al. (2023a)²⁵ explored the N_2O emissions from anammox-based processes. They revealed the relationship between N_2O emissions and operational factors as well as microbial communities through machine learning models and corresponding interpretable analysis. These studies have demonstrated that machine learning models have performed well in big data analysis for anammox-based processes.^{23–25}

However, the response effects of nitrogen treatment performance in the different types of anammox-based processes to multiple operational factors are still unclear. Moreover, previous studies for big data analysis of anammox-based processes established machine learning models manually^{23–25}, which suffers from several drawbacks. On one hand, the procedure of training and tuning these machine learning models had a significant time cost²². On the other hand, the accuracy of these trained models heavily depended on the experience of model designers, which had poor reproducibility and robustness. To overcome these drawbacks, automated machine learning (AutoML) algorithms have been proposed and drawn increasing attention^{22,26,27}.

Therefore, in this study, the AutoML algorithm was applied to predict nitrogen treatment performance and interpret the response effects of anammox-based processes to multiple operational factors. Data from relevant anammox-based studies were first analyzed to select the potential input and output variables, which formed the integrated dataset for running the H₂O AutoML algorithm. Then, the candidate models generated by the H₂O AutoML algorithm for 7 output variables were compared. Moreover, lab-scale anammox-based experiments were carried out to test the prediction performance of candidate models. Based on the optimal models, interpretable analysis was conducted to clarify the response effects of output variables to the important input variables. These results highlight previously overlooked yet crucial relationships impacting the anammox-based treatment performance, providing a significant step towards the improvement of mainstream anammox-based processes.

Materials and methods

Data sources and analysis methods

To reveal internal relationships among factors in anammox-based nitrogen removal processes, this study first conducted a comprehensive literature search on the Web of Science using keywords such as “Anaerobic ammonia oxidation”, “mainstream anammox”, “municipal wastewater”, “partial nitrification”, and “partial denitrification”. The content of this literature was examined and 28 studies were finally selected because they had comprehensive and consistent variables related to the anammox-based nitrogen removal processes (Table S9). Liu et al. (2020)²⁸ conducted a meta-analysis for anammox-based processes with data collected from 62 studies. However, they did not consider the operational parameters such as hydraulic retention time (HRT) and nitrogen loading rate (NLR) which were utilized as input variables in our study. In total, 2940 samples from different types of anammox-based processes in the selected study were collected by the Origin software, including 1927 PDA samples, 282 PNA samples, and 731 PDA combined with PNA samples.

The key variables involved in these anammox-based processes were organized, including operation condition (sequencing batch OR continuous flow), sludge morphology (floc OR aggregates), influent type (synthetic wastewater OR municipal wastewater), enrichment strategy (self-enrichment OR inoculation), anammox-based process type (PNA OR PDA OR PNA combined with PDA), operation time (d), HRT (h), carbon to nitrogen ratio (C/N), influent chemical oxygen demand (COD, mg/L), influent NH_4^+ -N (mg/L), influent NO_3^- -N (mg/L), influent NO_2^- -N (mg/L), influent total inorganic nitrogen (TIN, mg/L), effluent NH_4^+ -N (mg/L), effluent NO_3^- -N (mg/L), effluent NO_2^- -N (mg/L), effluent TIN (mg/L), and genera of dominant anammox bacteria (Table S1). Furthermore, several mathematic indices including NLR ($\text{kg}/\text{m}^3/\text{d}$), NH_4^+ -N removal efficiency (%), TIN removal efficiency (%), nitrification reaction rate by the ammonia oxidation bacteria (N_RR), nitrogen removal rate through the anammox reaction pathway ($\text{N}_\text{A,RR}$), nitrogen removal rate (NRR) were calculated according to the equations in the Supporting information (Text S1)²⁵. To preliminary analyze the relationships among these variables, box plots, exploratory data analysis (EDA), and Pearson product-moment correlation coefficient (PPMC) were utilized according to our previous study²².

Model development and optimization

In total, 24 variables were collected from relevant literature to be selected as the input and output variables of regression machine learning models (Fig. 1). Among them, the operation condition, sludge morphology, influent type, enrichment strategy, anammox-based process type and genera of dominant anammox bacteria were represented as categorical objects in the dataset (Table S1). The optional 15 input variables included operation condition, sludge morphology, influent type, enrichment strategy, process type, dominant anammox bacteria, operation time, HRT, C/N, influent COD, influent $\text{NH}_4^+ \text{-N}$, influent $\text{NO}_3^- \text{-N}$, influent $\text{NO}_2^- \text{-N}$, influent TIN, and NLR. The 7 output variables including effluent $\text{NH}_4^+ \text{-N}$, effluent $\text{NO}_3^- \text{-N}$, effluent $\text{NO}_2^- \text{-N}$, effluent TIN, $\text{NH}_4^+ \text{-N}$ removal efficiency, TIN removal efficiency, and N_A RR were predicted separately to identify the relationships between input variables and the specific output variable. The selection of input variables was according to PPMC results.

To reduce prediction errors and avoid non-convergence of data, 2940 samples were randomly selected and divided into the training dataset (70%), validation dataset (15%), and testing dataset (15%) (Fig. 1). As a result, the training sample-size to the feature-size ratio (SFR) reached 138 (2070 training samples divided by

Chúng tôi tuân theo các quy trình được công bố trước công chúng trước đây của chúng tôi22 để thiết lập các mô hình bằng nền tảng H2O AutoML (Text S2). Tóm lại, hình thuật toán học máy khác nhau bao gồm rừng ngẫu nhiên phân tán (DRF), cây ngẫu nhiên cực độ (XRT), Mô hình Tuyến tính Khai thác học để tăng cường, tăng cường độ dốc (Gradient Boosting) cực độ (XGBoost), mô hình tăng cường độ dốc (GBM), và các mạng nơ-ron sâu đã được tối ưu hóa và so sánh để tạo thành bảng xếp hạng mô hình (Fig. 1). Hơn nữa, sự kết hợp giữa tìm kiếm ngẫu nhiên nhanh và kiểm định chéo 5 lần (fivefold cross-validation) đã được áp dụng để tăng tốc độ huấn luyện và nâng cao hiệu suất của mô hình22. Việc tính chính phủ học cho nhiều loại mô hình học máy vốn rất khó khăn và tiêu tốn nhiều thời gian. Nền tảng H2O AutoML được lựa chọn vì có hiệu quả cao trong cả phát triển mô hình tự động lẫn giải thích mô hình30. Nhờ hiệu quả của các nền tảng H2O AutoML, số lượng mô hình tối đa đã được đặt là 200 và thời gian chạy tối đa đã được đặt là 900 s. Quá trình chạy AutoML sẽ dừng ngay khi đạt đến một trong hai giới hạn này. Năm giá trị seed khác nhau đã được sử dụng cho các lần phân chia dữ liệu khác nhau vì việc phân chia dữ liệu có thể dẫn đến sự khác biệt lớn về hiệu suất dự đoán cuối cùng.9. Hiệu suất huấn luyện, kiểm định và thử nghiệm tốt nhất của các mô hình ứng viên này đã được tóm tắt dựa trên sự so sánh độ trung bình (MAE) và Hệ số xác định (R2).

Fig. 1. Sơ đồ quy trình của phương pháp học máy tự động nhằm phân tích dữ liệu lớn thu thập từ các quy trình loại bỏ nitơ dựa trên quá trình anammox.

Phân tích khả năng diễn giải

Các phương pháp điển giải bao gồm từ quan trọng của biến, biểu đồ phụ thuộc riêng phần (PDP) một chiều (1D PDP), hai chiều (2D PDP) được sử dụng để phân tích các mô hình ưu tiên tối ưu để thuật toán học máy tự động (AutoML) lựa chọn cho từng biến đầu ra. Tâm quan trọng của biến được dùng để tính toán mức độ quan trọng của các đặc trưng mẫu và mô tả định lượng đóng góp của các biến trung gian cho phân loại hoặc hồi quy. PDP thể hiện mối quan hệ hàm của một hoặc hai biến đặc trưng (1D PDP hoặc 2D PDP) liên kết với giá trị đoán của mô hình. Trong 2D PDP, giá trị của hai biến đặc trưng được chọn thay đổi đồng thời, trong khi giá trị của các đặc trưng khác được giữ cố định. Kết quả dự đoán sau đó được tính toán nhằm thể hiện ảnh hưởng kết hợp của hai biến đặc trưng lên kết quả đầu ra. Phần tích khả năng diễn giải được thực hiện trong Flow UI của nền tảng học máy tự động H2O (H2O AutoML) (<https://h2o-release.s3.amazonaws.com/h2o/rel-3.46.0/6/docs-website/h2o-decs/flow.html>).

Kết quả và thảo luận

Hơn nữa, NIRR, NARR và NRR có mối tương quan mạnh với các biến liên quan đến công thức tính toán của chúng (Text S1). Đồng thời, NH4-N đóng vào và TIN đóng vào có mối quan hệ tương cấp rất tương đồng với các biến khác của NH4-N chiếm tỷ lệ cực kỳ cao ở các nước thải nhân tạo và hệ thống xử lý nước thải nhân tạo (Text S2). Như vậy, nên chọn biến liên quan đến công thức tính toán để được minh họa ở Fig. S3. Các mối quan hệ từng cặp mạnh mẽ đã được tìm thấy giữa lưu lượng thải đầu vào và vi khuẩn anammox từ thể, NO3-N đóng vào cùng chỉ số tín hiệu đầu vào. Mối tương quan này làm nổi bật hơn nữa sự khác biệt giữa nước thải nhân tạo và nước thải đô thị. Ngoài trừ NH4-N đóng vào và TIN đóng vào, không có mối quan hệ tuyến tính mạnh nào khác được tìm thấy giữa các biến đầu vào vào tiềm năng đó. Do đó, các biến này có thể được sử dụng làm các biến đầu vào để xây dựng mô hình học máy vi chứng không có nguy cơ đã công ty tiềm ẩn 19. Mối quan hệ giữa các biến này trong các quá trình loại bỏ nitơ dựa trên anammox cần được khám phá sâu hơn thông qua các mô hình để hiểu thêm về điều và các phương pháp giải thích được.

Fig. 2. Phân tích thống kê các quá trình loại bỏ nitơ dựa trên anammox. (a) Biểu đồ hộp về các tính chất dòng ra (I, □) các thí nghiệm með theo chu kỳ (sequencing batch) và (□, IV) các thí nghiệm dòng chảy liên tục (continuous flow); (b) Biểu đồ hộp về các tính chất dòng ra (I, □) các thí nghiệm bong bóng (floc) và (□, IV) các thí nghiệm tập hợp hạt (aggregates); (c) Phân tích dữ liệu kém (EDA) đối với các tính chất dòng vào của nước thải nhân tạo; (d) Phân tích dữ liệu khám phá (EDA) đối với các tính chất dòng vào của nước thải đô thị; (e) Hệ số tương quan tích Pearson (PPMC) cho các thí nghiệm PNA; (f) Hệ số tương quan tích Pearson (PPMC) cho các thí nghiệm PDA; (g) Hệ số tương quan tích Pearson (PPMC) cho các thí nghiệm PNA kết hợp với PDA. NLR: tải lượng nạp nitơ; NIRr: tốc độ phản ứng nitrit hóa bởi vi khuẩn oxy hóa amoni; NARR: tốc độ xử lý nitơ qua con đường phản ứng anammox; NRR: tốc độ loại bỏ nitơ.

Mặt khác, một số biến cơ chế còn thiếu trong tập dữ liệu, chẳng hạn như pH, DO và các chất hữu cơ gây ức chế đặc thù có thể là những nguyên nhân dẫn đến độ chính xác thấp đối với NO_2^- -N dòng ra và tốc độ xử lý nitơ qua anammox (NARR).

Bảng 2. Hiệu năng tốt nhất và các mô hình tối ưu được tạo ra bởi thuật toán H2O AutoML cho tập dữ liệu thực nghiệm chưa từng thấy.

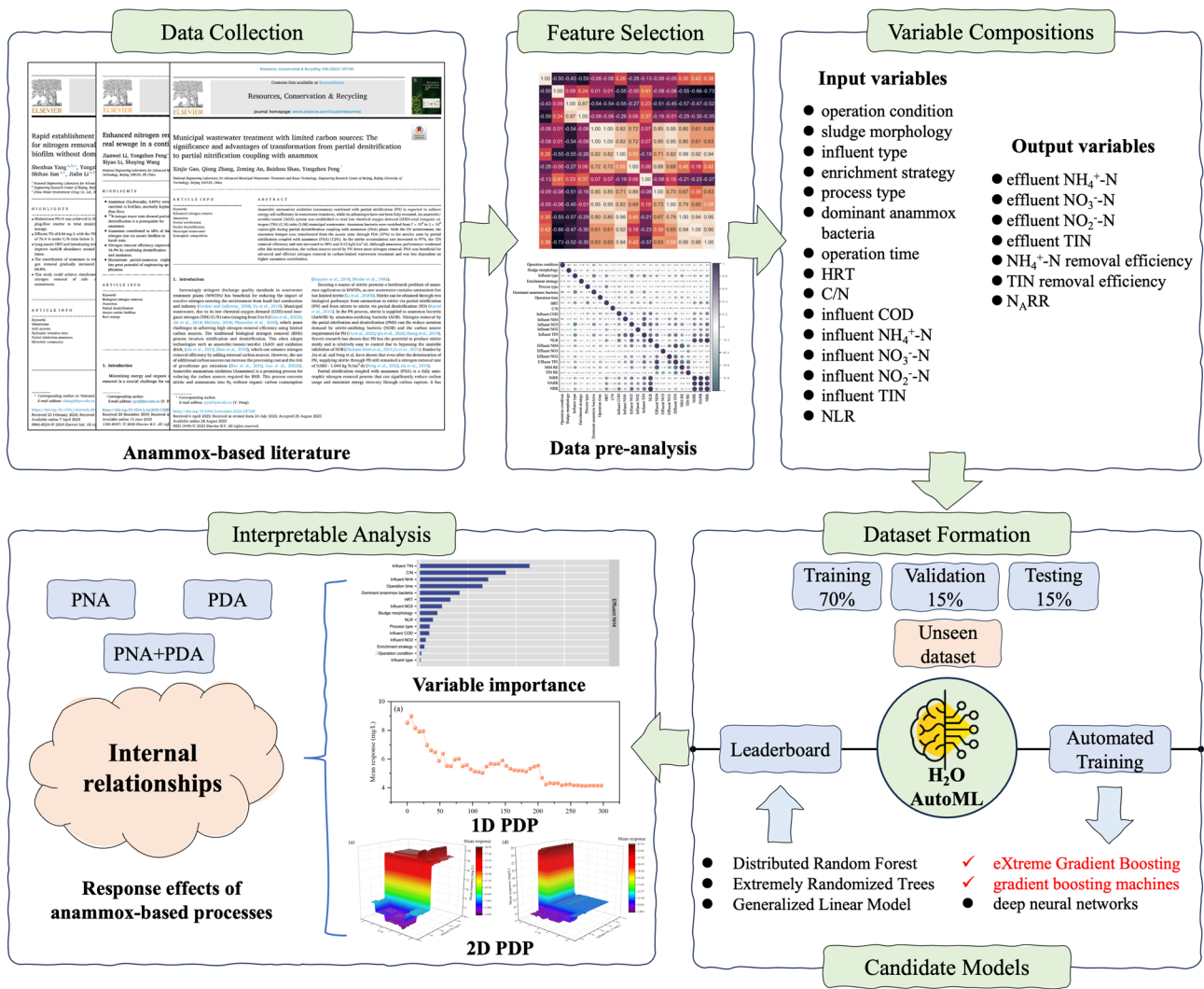


Fig. 1. Flowchart of the automated machine learning approach to analyze the big data collected from anammox-based nitrogen removal processes.

a total of 15 input variables), which presented higher confidence and good learning potential ($\text{SFR} \geq 100$)^{19,29}. Standardization (mean and standard deviation) for numeric columns was embedded in the H₂O AutoML platform. To improve the model efficiency, correlation analysis was conducted to remove features that may cause potential multicollinearity, and retain features highly correlated with the output variable (Fig. S1)²⁹. We followed the procedures introduced in our previous work²² to establish models by the H₂O AutoML platform (Text S2). Briefly, various machine learning algorithms including Distributed Random Forest (DRF), Extremely Randomized Trees (XRT), Generalized Linear Model with regularization, eXtreme Gradient Boosting (XGBoost), gradient boosting machines (GBM), and deep neural networks were automatically optimized and compared to form the modeling leaderboard (Fig. 1). Moreover, the combination of fast random search and fivefold cross-validation was applied to accelerate the training speed and enhance the model performance²². The proper tuning of numerous kinds of machine learning models is notoriously difficult and time-consuming¹⁹. The H₂O AutoML platform was selected because it has high efficiency in both automated model development and model explanation³⁰. Due to the high efficiency of the H₂O AutoML platform, the maximum number of models was set as 200 and the maximum run time was set as 900 s. AutoML run will stop as soon as it hits either of these limits. Five different seeds were utilized for different data splitting because the data splitting may result in a huge discrepancy in the final prediction performance¹⁹. The best training, validation, and test performance of these candidate models were summarized based on mean absolute error (MAE) and Coefficient of Determination (R^2).

Experimental validation of candidate models

To confirm the high performance of candidate models, the candidate models including DRF, XRT, XGBoost, GBM, Generalized Linear Model, and deep neural networks trained by the same training dataset (2070 training samples) were all tested by the unseen experimental dataset (185 samples). A PDA up-flow anaerobic sludge

Bảng 1. Hiệu năng tốt nhất và các mô hình tối ưu được tạo ra bởi thuật toán học máy tự động (AutoML) H₂O với năm seed phân chia dữ liệu.

Nhìn chung, độ chính xác dự đoán cao của các mô hình đã huấn luyện này đối với dữ liệu thực nghiệm chưa từng thấy cho thấy các mô hình ứng viên do nền tảng học máy tự động (AutoML) H₂O tạo ra có khả năng khái quát hóa xuất sắc trong việc dự đoán hiệu năng của các quá trình loại bỏ nitơ dựa trên anammox¹⁹.

bed (UASB) reactor with a 5.72 L working volume was established to investigate the PDA performance under different C/N. The PD flocculent sludge from a long-term operating glycerol-driven PD reactor and the mature anammox granular sludge from a UASB anammox reactor were mixed with the biomass ratio of 1:5 and inoculated into the PDA UASB reactor. Synthetic wastewater was prepared according to our previous study³¹. The reaction temperature was kept at 35 °C using the water bath, and the HRT was set at 4 h. The categorical variables of this experiment have been summarized in Table S9. The PDA UASB reactor operated for 185 d with 6 stages under different influent conditions (Table S10). The varying treatment performance of this experiment including effluent $\text{NH}_4^+\text{-N}$, effluent $\text{NO}_3^-\text{-N}$, effluent $\text{NO}_2^-\text{-N}$, effluent TIN, $\text{NH}_4^+\text{-N}$ removal efficiency, TIN removal efficiency, and N_A RR is illustrated in Fig. 5. These nitrogen concentrations of samples were measured according to the Standard Methods³².

Interpretable analysis

Interpretable methods including variable importance, one-dimensional (1D), and two-dimensional (2D) partial dependence plot (PDP) were utilized to analyze the optimal candidate models generated by the AutoML algorithm for each output variable. Variable importance is used to calculate the importance of sample features and quantitatively describe the contribution of features to classification or regression³³. The PDP can show the functional relationship of one or two characteristic variables (1D PDP or 2D PDP) on the model's prediction results³⁴. In 2D PDP, the values of the two selected features are changed at the same time, while the values of the other features are fixed, and the prediction results are calculated to show the combined effect of the two features on the prediction results. The interpretable analysis was conducted on the Flow UI of H₂O AutoML platform (<https://h2o-release.s3.amazonaws.com/h2o/rel-3.46.0/6/docs-website/h2o-docs/flow.html>).

Results and discussion

Statistical analysis of anammox-based nitrogen removal processes

To better understand the characteristics of data collected from the anammox-based processes, the box plots, EDA, and PPMC were utilized to analyze collected data (Fig. 2). There were 2940 samples collected from 28 studies, which investigated different types of anammox-based processes and applied different operation conditions. Box plots illustrated the distribution of data from different groups. Outliers in all groups indicated occasional significant deviations among different experiments. As shown in Fig. 2(a), the sequencing batch experiments (median = 78.18%) showed a higher TIN removal efficiency than the continuous flow experiments (median = 74.81%). Meanwhile, more stable NLR, N_I RR, N_A RR, and NRR were observed in the sequencing batch experiments. Extremely high NLR (2.91–7.55 kg N/m³/d), Ni RR (0.33–2.53 kg N/m³/d), N_A RR (0.02–3.92 kg N/m³/d) and NRR (2.63–6.55 kg N/m³/d) were found in Du et al. (2016)³⁵ and Wang et al. (2024b)³⁶ due to their high influent nitrogen concentration and short HRT in the UASB and expanded granular sludge bed reactor operating under continuous flow. This difference was also obtained between the floc experiments and aggregate experiments (Fig. 2(b)). The high NLR in continuous flow reactors operated by aggregates demonstrated that this combination had great potential to treat wastewater with high levels of nitrogen^{37,38}.

For experiments using synthetic wastewater and municipal wastewater, the effluent properties were similar (Fig. S1(a)). However, the experiments using municipal wastewater had more stable NLR, N_I RR, N_A RR, and NRR (Fig. S1(b)), suggesting that influent properties of municipal were analogous across different studies^{39–42}. Instead, the synthetic wastewater utilized in different studies had a larger range of influent COD, $\text{NH}_4^+\text{-N}$ and $\text{NO}_3^-\text{-N}$ concentrations^{35,36,43,44}. To further analyze the differences in influent properties of synthetic and municipal wastewater, the distribution of influent properties was visualized by the EDA approach (Fig. 2(c) and (d)). Data points in larger regions were observed in the C/N versus influent $\text{NH}_4^+\text{-N}$, C/N versus influent TIN, influent COD versus influent $\text{NH}_4^+\text{-N}$, and influent COD versus influent TIN in Fig. 2(d), indicating that the influent variables of synthetic wastewater were more concentrated and had less flocculation than that of municipal wastewater. In reality, the concentrations of pollutants in real wastewater normally fluctuate greatly^{45,46}. Therefore, the experiments using synthetic wastewater need to further take into account the real composition and fluctuation of the municipal wastewater.

Notably, the data from different anammox-based process types (i.e., PNA, PDA, and PNA combined with PDA) presented high differences (Fig. S2). The PNA and PNA combined with PDA experiments had higher TIN removal efficiency than the PDA experiments (Fig. S2(a)). In all three types of anammox-based processes, the effluent $\text{NH}_4^+\text{-N}$, effluent $\text{NO}_2^-\text{-N}$, and effluent TIN were positively correlated to the influent $\text{NH}_4^+\text{-N}$, influent TIN, and NLR, whereas the strongest positive correlation (0.55–0.95) was found in the PNA experiments (Fig. 2(e), (f) and (g)). Unlike PDA processes, the realization of PNA processes requires inhibition of nitrite-oxidizing bacteria (NOB) activity and enhancement of ammonia-oxidizing bacteria (AOB) activity⁴⁷. The NOB and AOB activity have been found to be largely affected by the substrate concentrations (e.g., $\text{NO}_2^-\text{-N}$, $\text{NH}_4^+\text{-N}$, and COD)^{48,49}. This may be the reason for the stronger correlation between influent and effluent variables in PNA experiments.

Moreover, the N_I RR, N_A RR, and NRR had strong correlations with the variables involved in their calculations (Text S1). Meanwhile, the influent $\text{NH}_4^+\text{-N}$ and influent TIN had very similar pairwise relationships with other variables because $\text{NH}_4^+\text{-N}$ was extremely high in both synthetic and municipal wastewater. For real municipal wastewater, the $\text{NH}_4^+\text{-N}$ normally accounts for more than 90% of influent TIN^{42,50}.

The PPMC for all categorical and numerical variables in the entire dataset is illustrated in Fig. S3. Strong pairwise relationships were found between the influent type and dominant anammox bacteria, influent $\text{NO}_3^-\text{-N}$ as well as influent TIN. This correlation further highlighted the differences between synthetic wastewater and municipal wastewater. Except for influent $\text{NH}_4^+\text{-N}$ and influent TIN, no other strong linear relationship was found between these potential input variables. Therefore, these variables could be utilized as the input variables for establishing machine learning models because they lacked potential multicollinearity¹⁹. The relationships

Fig. 3. Tầm quan trọng của 15 biến đầu vào đối với 9 biến đầu ra. (a) $\text{NH}_4^+\text{-N}$ dòng ra; (b) $\text{NO}_3^-\text{-N}$ dòng ra; (c) $\text{NO}_2^-\text{-N}$ dòng ra; (d) tổng nitơ vô cơ (TIN) dòng ra; (e) hiệu suất loại bỏ $\text{NH}_4^+\text{-N}$; (f) hiệu suất loại bỏ tổng nitơ vô cơ (TIN); (g) tốc độ xử lý nitơ qua anammox (NARR).

Hơn nữa, thời gian vận hành cũng có ảnh hưởng tương hỗ tương tự với TIN dòng vào (Fig. 4(c)) và $\text{NH}_4^+\text{-N}$ dòng vào (Fig. 4(f)) lên $\text{NH}_4^+\text{-N}$ dòng ra. Đáp ứng giảm đáng kể của $\text{NH}_4^+\text{-N}$ dòng ra từ 32.16 mg/L xuống 4.68 mg/L và từ 20.46 mg/L xuống 4.12 mg/L được ghi nhận tương ứng đối với thời gian vận hành so với TIN dòng vào và thời gian vận hành so với $\text{NH}_4^+\text{-N}$ dòng vào. Đáng chú ý, 2D PDP của thời gian vận hành so với C/N có các đỉnh và đáy phức tạp nhất (Fig. 4(e)). Đáp ứng thấp của $\text{NH}_4^+\text{-N}$ dòng ra được tìm thấy ở tỷ lệ C/N từ 2.75–6.48, phù hợp với các kết quả của 1D PDP.

Ngoài ra, thời gian vận hành và các chỉ vi khuẩn anammox ưu thế đóng vai trò quan trọng thứ hai trong việc dự đoán loại bỏ $\text{NH}_4^+\text{-N}$, ngụ ý rằng sự tiến hóa của các quần xã vi sinh vật và sự làm giàu của các chi anammox trong suốt các thí nghiệm có vai trò quyết định đối với việc tiêu thụ $\text{NH}_4^+\text{-N}$ trong các quá trình dựa trên anammox. Thử ví là, hình thái bùn, loại quy trình, chiến lược làm giàu vi sinh, điều kiện vận hành và loại dòng vào không thể hiện tầm quan trọng đáng kể trong việc dự đoán loại bỏ $\text{NH}_4^+\text{-N}$ (Fig. 3(a) và (e)). Những kết quả này có thể do các con đường loại bỏ nitơ tương tự nhau trong các quá trình loại bỏ nitơ dựa trên anammox với các hình thái bùn, loại quy trình, chiến lược làm giàu vi sinh và điều kiện vận hành khác nhau. Loại dòng vào là biến ít quan trọng nhất để dự đoán loại bỏ $\text{NH}_4^+\text{-N}$, cho thấy sự khác biệt giữa nước thải nhân tạo và nước thải đô thị có thể được bỏ qua trong các quá trình loại bỏ $\text{NH}_4^+\text{-N}$ (Fig. 5).

Fig. 4. Biểu đồ phụ thuộc riêng phần hai chiều (2D PDP) về tương tác giữa hai biến đầu vào để dự đoán $\text{NH}_4^+\text{-N}$ dòng ra. (a) TIN dòng vào so với C/N; (b) TIN dòng vào so với $\text{NH}_4^+\text{-N}$ dòng vào; (c) TIN dòng vào so với thời gian vận hành; (d) C/N so với $\text{NH}_4^+\text{-N}$ dòng vào; (e) C/N so với thời gian vận hành; (f) thời gian vận hành so với $\text{NH}_4^+\text{-N}$ dòng vào.

Đối với hiệu suất loại bỏ TIN, PDP hai chiều (2D PDP) của thời gian vận hành so với TIN dòng vào (Fig. S12(a)), COD dòng vào (Fig. S12(b)), C/N (Fig. S12(c)), HRT (Fig. S12(d)) và NLR (Fig. S12(e)) thể hiện rõ đáp ứng ngày càng tăng

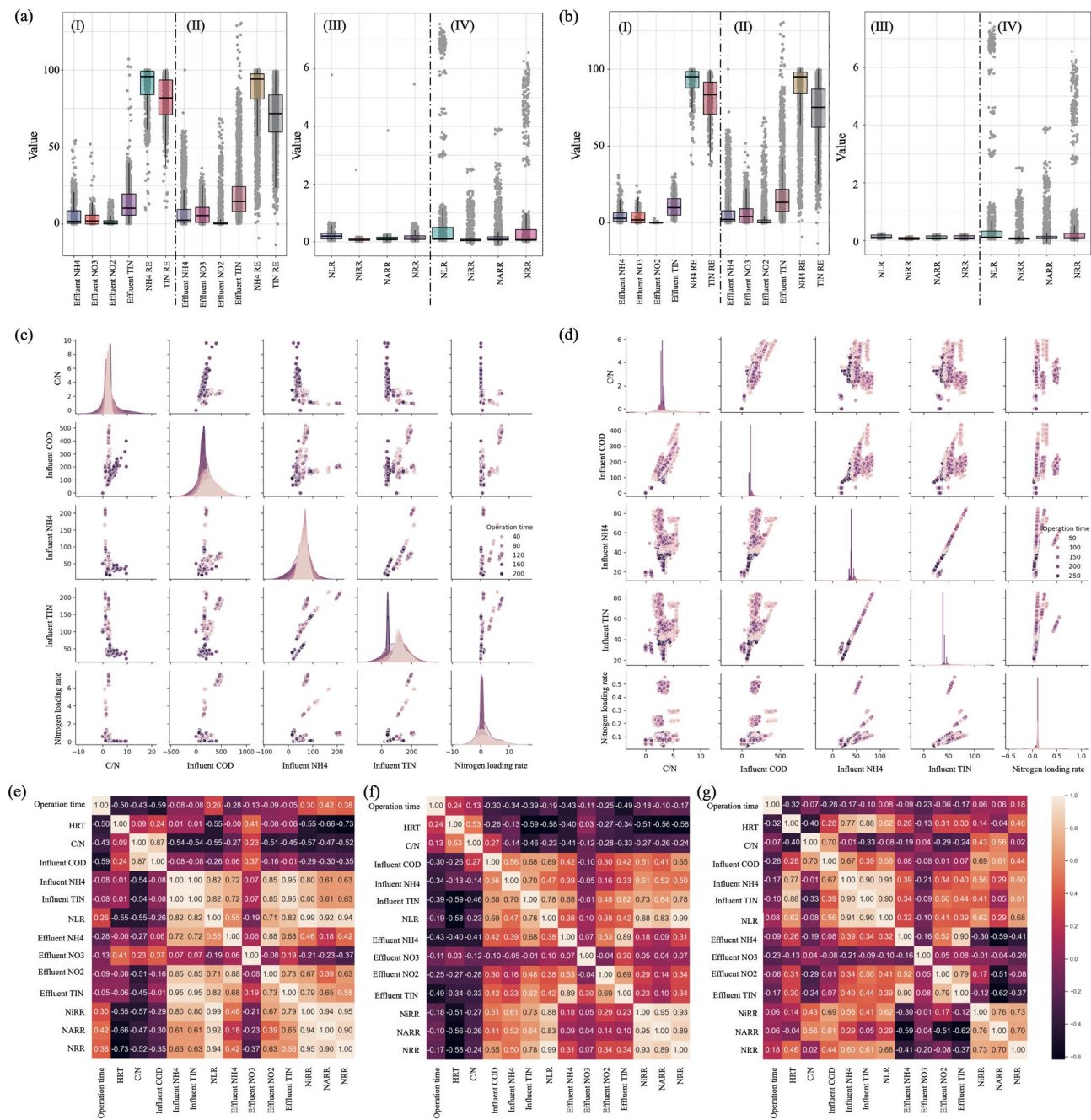


Fig. 2. Statistical analysis of anammox-based nitrogen removal processes. (a) the box plot of effluent properties from (I, III) sequencing batch experiments and (II, IV) continuous flow experiments; (b) the box plot of effluent prosperities from (I, III) flocculent experiments and (II, IV) aggregate experiments; (c) Exploratory Data Analysis for influent properties of synthetic wastewater; (d) Exploratory Data Analysis for influent properties of municipal wastewater; (e) Pearson product-moment correlation coefficient for PNA experiments; (f) Pearson product-moment correlation coefficient for PDA experiments; (g) Pearson product-moment correlation coefficient for PDA combined with ammonia combined with PDA experiments. NLR: nitrogen loading rate; N_{RR} : nitrification reaction rate by the ammonia oxidation bacteria; N_{ARR} : nitrogen removal rate through the anammox reaction pathway; NRR: nitrogen removal rate.

Predicted target	Optimized model	Training MAE	Training R ²	Validation MAE	Validation R ²	Testing MAE	Testing R ²
Effluent NH ₄ ⁺ -N	GBM	0.171	0.994	1.951	0.914	1.889	0.928
Effluent NO ₃ ⁻ -N	XGBoost	0.048	0.998	1.677	0.727	1.626	0.824
Effluent NO ₂ ⁻ -N	XGBoost	0.004	0.999	0.546	0.949	0.496	0.962
Effluent TIN	XGBoost	0.215	0.996	3.141	0.916	3.366	0.910
NH ₄ ⁺ -N removal efficiency	XGBoost	0.305	0.996	3.638	0.832	3.975	0.882
TIN removal efficiency	XGBoost	0.320	0.991	5.583	0.731	5.252	0.814
N _A RR	XGBoost	0.002	0.999	0.013	0.981	0.014	0.993

Table 1. The best performance and optimized models generated by the H₂O AutoML algorithm with five data splitting seeds.

Predicted target	Optimized model	MAE	R ²
Effluent NH ₄ ⁺ -N	GBM	1.686	0.945
Effluent NO ₃ ⁻ -N	XGBoost	2.196	0.725
Effluent NO ₂ ⁻ -N	GBM	1.045	0.563
Effluent TIN	GBM	3.407	0.899
NH ₄ ⁺ -N removal efficiency	GBM	4.514	0.867
TIN removal efficiency	GBM	3.375	0.882
N _A RR	Deep learning	0.013	0.677

Table 2. The best performance and optimized models generated by the H₂O AutoML algorithm for the unseen experimental dataset.

among these variables involved in anammox-based nitrogen removal processes should be further explored via data-driven models and interpretable methods.

Predicting nitrogen removal performance in literature-based anammox processes

Based on the above pairwise correlation analysis (Fig. S3), all 15 variables including 6 categorical variables and 9 numerical variables were selected as the input variables for predicting the effluent NH₄⁺-N, effluent NO₃⁻-N, effluent NO₂⁻-N, effluent TIN, NH₄⁺-N removal efficiency, TIN removal efficiency, and N_ARR, separately.

The best performance and optimized models for each output variable are summarized in Table 1 and the Supporting Information. GBM-like models (i.e., GBM and XGBoost) presented the advantages in predicting effluent NH₄⁺-N, effluent NO₃⁻-N, effluent NO₂⁻-N, effluent TIN, NH₄⁺-N removal efficiency, TIN removal efficiency, and N_ARR. GBM has been recognized as an ensemble and powerful supervised machine learning algorithm^{22,51}. GBM demonstrates remarkable performance advantages owing to its integration of regularization techniques that effectively prevent overfitting. It also features built-in handling of missing values, efficient tree-pruning algorithms, and parallel processing capabilities. Additionally, it employs advanced methods for addressing imbalanced data. Collectively, these features enhance the model's efficiency, accuracy, and robustness when dealing with complex data scenarios, making it an excellent choice for a wide range of machine learning tasks⁵². For instance, GBM also presented high prediction performance in simulating total nitrogen in the influent of WWTPs⁵³ and energy consumption of WWTPs⁵⁴.

The low training MAE and high training R² (0.991–0.999) of these models demonstrated that the big dataset collected from literature-based anammox processes was useful for training machine learning models. The validation performance was weaker than the training performance, but it still fulfilled the prediction requirement (R²=0.727–0.981). Notably, the testing performance of these models (R²=0.814–0.993) was better than the validation performance because the modeling leaderboard was sorted based on the testing performance.

Testing candidate models by the experimental dataset

To confirm the high performance of candidate models generated by the H₂O AutoML platform, the candidate models trained by the same training dataset (2070 training samples) were tested by the unseen experimental dataset (185 samples). The optimized models for predicting 7 output variables are summarized in Table 2. High prediction accuracy (R²=0.725–0.945) was achieved for effluent NH₄⁺-N (Fig. S15), effluent NO₃⁻-N (Fig. S16), effluent TIN (Fig. S18), NH₄⁺-N removal efficiency (Fig. S19) and TIN removal efficiency (Fig. S20). Although the prediction accuracy of effluent NO₂⁻-N and N_ARR was lower than other output variables, the predicted values of effluent NO₂⁻-N (Fig. S17) and N_ARR (Fig. S21) still fulfilled the overall trends of real values in experiments (Fig. 5). The prediction MAE of effluent NO₂⁻-N (1.045) by the GBM was still lower than the MAE of ensemble regression trees (3.428) created by Huang et al. (2023)⁵⁵ based on the input variables including operating days, influent NH₄⁺-N, influent NO₂⁻-N, effluent pH, and effluent DO. The low prediction accuracy of the effluent NO₂⁻-N and N_ARR may be attributed to the high flocculation of these values in the anammox-based experiments (Fig. 5(c) and (g)). Moreover, the N_ARR was calculated based on multiple variables (Text S1), which brought high uncertainty in it. On the other hand, some missing mechanistic variables in dataset, such as

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Thứ hai, nhiều chất hữu cơ với cấu trúc hóa học phức tạp cùng tồn tại trong nước thải đô thị. Trong số đó, một số nhóm thế (ví dụ: -CH3, -OH) trong các chất hữu cơ tác động đến sự khuếch tán cơ chất và độ ổn định màng thông qua việc thay đổi tính kỵ nước. Hơn nữa, một số nhóm thế (ví dụ: -Cl, -F) trong các chất hữu cơ góp phần gây độc sinh học và sẽ ức chế hoạt tính anammox⁷². Tuy nhiên, COD lại được dùng làm chỉ số đại diện cho toàn bộ các chất hữu cơ này trong phần lớn các thí nghiệm anammox sử dụng nước thải đô thị. Việc thiếu hụt nồng độ của các chất hữu cơ nhạy cảm này trong tập dữ liệu sẽ hạn chế khả năng của các mô hình học máy trong việc nắm bắt ảnh hưởng của các chất hữu cơ khác nhau đến các quá trình loại

Tính sẵn có của dữ liệu Kết luận

Tất cả dữ liệu hỗ trợ cho các phát hiện của nghiên cứu này đều có sẵn trong bài báo và phần Thông tin bổ sung (Supplementary Information).

Nhận bài: 5 tháng 3 năm 2026; Chấp nhận đăng: 4 tháng 6 năm 2026

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pH, DO and specific inhibitory organic substances, may be the reasons for the low accuracy of effluent NO_2^- -N and $\text{N}_\text{A}\text{RR}$.

Overall, the high prediction accuracy of these trained models for unseen experimental data indicated that the candidate models generated by the H_2O AutoML platform have excellent generalization ability in predicting the performance of anammox-based nitrogen removal processes¹⁹.

Interpretable analysis of the optimized models for anammox-based nitrogen removal processes

To further understand the internal relationships between input variables and output variables, the variable importance of each input variable was analyzed for the optimized models (i.e., GBM and XGBoost mentioned above).

Effluent NH_4^+ -N

For effluent NH_4^+ -N and NH_4^+ -N removal efficiency, different variable importance was observed. The variable importance for predicting effluent NH_4^+ -N was found to follow the order of influent TIN > C/N > influent NH_4^+ -N > operation time > dominant anammox bacteria > HRT > influent NO_3^- -N > sludge morphology > NLR > process type > influent COD > influent NO_2^- -N > enrichment strategy > operation condition > influent type (Fig. 3(a)). Instead, the influent NO_3^- -N had higher importance for predicting NH_4^+ -N removal efficiency (Fig. 3(e)).

The variables (i.e., influent TIN, C/N, influent NH_4^+ -N, and influent NO_3^- -N) related to the carbon and nitrogen composition of the influent have been identified as the most important variables for predicting NH_4^+ -N variables, suggesting that the removal of NH_4^+ -N in anammox-based nitrogen removal processes were strongly affected by the detailed carbon and nitrogen compositions in influent. As indispensable substrate for the

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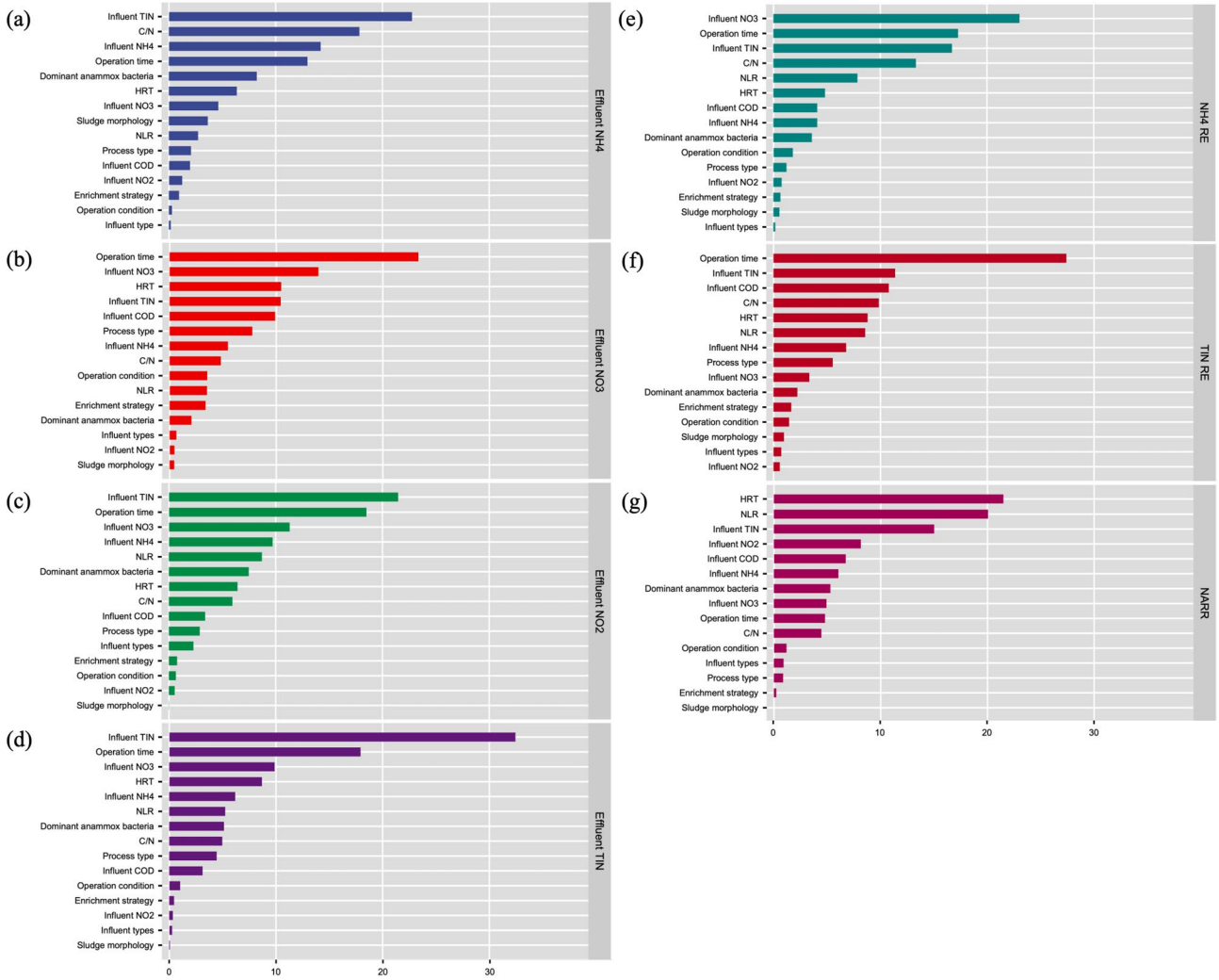


Fig. 3. The variable importance of 15 input variables for 9 output variables. (a) effluent NH_4^+ -N; (b) effluent NO_3^- -N; (c) effluent NO_2^- -N; (d) effluent TIN; (e) NH_4^+ -N removal efficiency; (f) TIN removal efficiency; (g) $\text{N}_\text{A}\text{RR}$.

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anammox processes, the NH_4^+ -N removal highly depends on the anammox activity, which is greatly influenced by the influent compositions^{56,57}. Meanwhile, the influent type (synthetic wastewater OR municipal wastewater) was the least important feature for predicting NH_4^+ -N removal. In most studies, the composition of municipal wastewater was also simplified as the basic pollutant concentrations (e.g., COD, NH_4^+ -N and TIN). Therefore, the difference between synthetic wastewater and municipal wastewater may be overlooked in the dataset, which may explain the least importance of influent type for predicting NH_4^+ -N removal.

To further explore the numerical dependence of NH_4^+ -N removal, the 1D PDP of important numerical input variables are illustrated (Fig. S4). The response of effluent NH_4^+ -N showed a decrease trend with the operation time (Fig. S4(a)), which agreed with the common observation of the improving treatment performance at the later period of anammox-based experiments^{14,58,59}. Meanwhile, the 1D PDP results revealed an appropriate range of C/N (2.72–6.32), in which lower effluent NH_4^+ -N concentrations were achieved compared with $C/N < 2.72$ and $C/N > 6.32$ (Fig. S4(b)). Previous studies also confirmed that too low or too high C/N led to the deteriorated NH_4^+ -N removal rates in the anammox processes^{60,61}, which was agreed with the 1D PDP results (Fig. S4(b)). For instance, Miao et al. (2018)⁶⁰ reported that the removal efficiency of NH_4^+ -N in the anammox process increased progressively as the C/N ratio rose from 1.1 to 2.5.

The 1D PDP of influent NH_4^+ -N and influent TIN has similar trend (Fig. S4(c) and (d)). A great increase of effluent NH_4^+ -N was found when the influent NH_4^+ -N and influent TIN reached 62.32 mg/L and 91.08 mg/L. These results suggested that municipal wastewater with nitrogen concentrations below these threshold values may be more suitable to be treated by the anammox-based nitrogen removal processes. Actually, the influent NH_4^+ -N and influent TIN in real municipal wastewater are normally below these threshold values^{62,63}, which ensured the high contribution of anammox process for municipal wastewater.

These input variables also have a mutual influence together on the effluent NH_4^+ -N. Therefore, the 2D PDP is illustrated to examine the influence of a couple of important numerical input variables on the effluent NH_4^+ -N as well as their mutual interaction (Fig. 4). Clear peaks and valleys were observed in the 2D PDP, implying that effluent NH_4^+ -N had strong dependence on these numerical input variables (Fig. 4). The lowest response of effluent NH_4^+ -N was achieved at the C/N of 2.75 and influent TIN from 18.85 mg/L to 87.76 mg/L (Fig. 4(a)). However, with the continuous increase of influent TIN, the response of effluent NH_4^+ -N extremely increased from 4.23 mg/L to 17.0 mg/L. At the same C/N, the increase of influent TIN leads to the increase of influent COD, which will cause the proliferation of heterotrophic bacteria to occupy the living space of anammox bacteria and inhibit its growth³⁶.

The 2D PDP of C/N versus influent NH_4^+ -N showed a similar trend to the 2D PDP of C/N versus influent TIN (Fig. 4(d)). However, a wider range of C/N (2.36–9.62) and influent NH_4^+ -N (13.83–213.84 mg/L) were observed with a low response of effluent NH_4^+ -N (7.17–10.74 mg/L). This difference between influent NH_4^+ -N and influent TIN may be due to the influent TIN composed of other types of nitrogen (e.g., NO_3^- -N, and NO_2^- -N). A notable increase response of effluent NH_4^+ -N on the increase of influent NH_4^+ -N and influent TIN was observed (Fig. 4(b)), which was agreed with the 1D PDP of influent NH_4^+ -N (Fig. S4(c)) and influent TIN (Fig. S4(d)).

Moreover, the operation time had similar mutual influence with the influent TIN (Fig. 4(c)) and influent NH_4^+ -N (Fig. 4(f)) on the effluent NH_4^+ -N. A greatly decreasing response of effluent NH_4^+ -N from 32.16 mg/L to 4.68 mg/L and from 20.46 mg/L to 4.12 mg/L was observed for operation time versus influent TIN and operation time versus influent NH_4^+ -N, respectively. Notably, the 2D PDP of operation time versus C/N had the most complex peaks and valleys (Fig. 4(e)). The low response of effluent NH_4^+ -N was found to be at the C/N of 2.75–6.48, which was consistent with the results of 1D PDP.

Besides, the operation time and genera of dominant anammox bacteria played second important roles in predicting NH_4^+ -N removal, implying that the evolution of microbial groups and enrichment of anammox genera during the experiments were crucial for consuming NH_4^+ -N in the anammox-based processes. Interestingly, the sludge morphology, process type, enrichment strategy, operation condition and influent type did not show strong importance in predicting NH_4^+ -N removal (Fig. 3(a) and (e)). These results may be due to the similar nitrogen removal pathways in the anammox-based nitrogen removal processes with different sludge morphology, process types, enrichment strategies and operation conditions. The influent type was the least important variable for predicting NH_4^+ -N removal, suggesting that the difference of synthetic and municipal wastewater could be overlooked in the NH_4^+ -N removal processes Fig. 5.

Effluent NO_3^- -N and NO_2^- -N

The operation time, influent NO_3^- -N, HRT, influent TIN, influent COD, and process type were the most important variables for predicting effluent NO_3^- -N (Fig. 3(b)), whereas the influent TIN, operation time, influent NO_3^- -N, influent NH_4^+ -N, NLR and dominant anammox bacteria were the most important variables for predicting effluent NO_2^- -N (Fig. 3(c)).

The 1D PDP of effluent NO_3^- -N and effluent NO_2^- -N response to these important variables is shown in Fig. S5 and S7. The operation time showed different effects on the effluent NO_3^- -N and effluent NO_2^- -N. A clear peak was observed at the operation time of 30 d for the effluent NO_3^- -N (Fig. S7(a)), then the response of effluent NO_2^- -N to operation time was similar to that of effluent NO_3^- -N. These results demonstrated that the NO_2^- -N generation was more than the NO_2^- -N consumption in the anammox-based nitrogen removal processes at the beginning of operation. In the mainstream anammox processes for municipal wastewater, NO_2^- -N is normally generated through PN or PD processes^{6,56}. At the beginning of operation, the unmatured anammox bacteria cannot consume the NO_2^- -N generated by the PN or PD processes. For instance, Yang et al. (2024)³⁹ reported that anammox only contributed 13.8% of nitrogen removal at the operation of 36–75 d, whereas its contribution increased to 67.1% at the operation of 216–258 d.

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Nguồn tài trợ

Tuyên bố

Xung đột lợi ích

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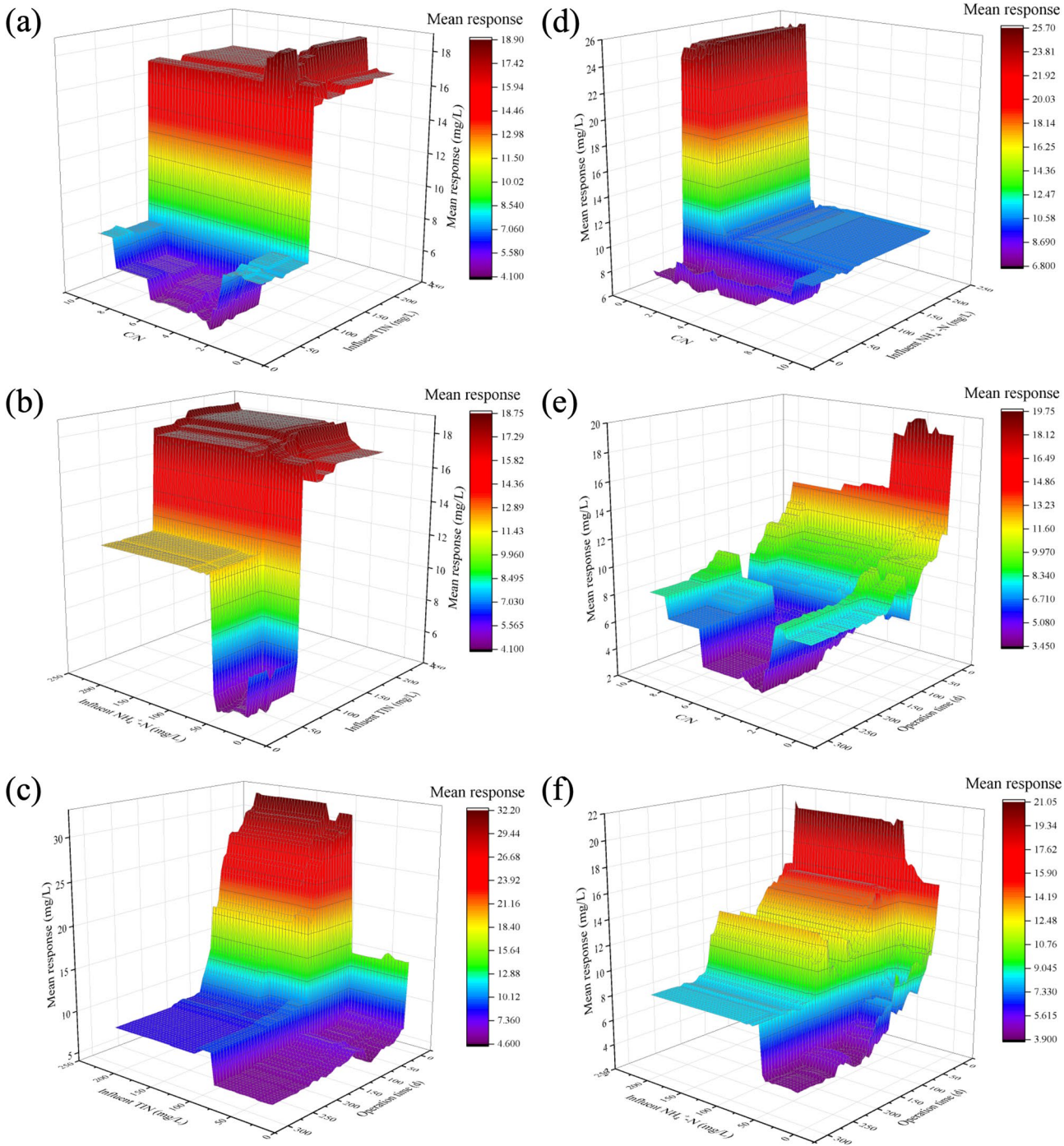


Fig. 4. 2D Partial dependence plot of two input variables interactions for predicting effluent $\text{NH}_4^+\text{-N}$. (a) influent TIN versus C/N; (b) influent TIN versus influent $\text{NH}_4^+\text{-N}$; (c) influent TIN versus operation time; (d) C/N versus influent $\text{NH}_4^+\text{-N}$; (e) C/N versus operation time; (f) operation time versus influent $\text{NH}_4^+\text{-N}$.

Interestingly, the response of effluent $\text{NO}_3^-\text{-N}$ had a strong peak and valley at the HRT of 10.47–15.13 h (Fig. S5(b)). Yang et al. (2024)³⁹ found that the HRT of 10 h did not improve the mainstream anammox processes, and further aggravated the issue of insufficient carbon sources. By increasing HRT to 17 h, the HRT of the anoxic zone increased from 5.67 h to 8.50 h. As a result, the average effluent $\text{NO}_3^-\text{-N}$ decreased from 14.46 mg/L to 9.79 mg/L³⁹. The prolongation of HRT affected the nitrite competition between denitrification and anammox processes, which had great effects on the effluent $\text{NO}_3^-\text{-N}$ ¹⁵. However, the effects of HRT should be considered combined with other factors such as influent COD, influent TIN, which decided the concentrations of substrates available to the denitrification and anammox processes.

The influent COD also strongly affected the effluent $\text{NO}_3^-\text{-N}$ (Fig. S5(c)). On one hand, low COD concentration (< 189.87 mg/L) may limit the heterotrophic denitrification activities³⁹, which causes the accumulation of $\text{NO}_3^-\text{-N}$

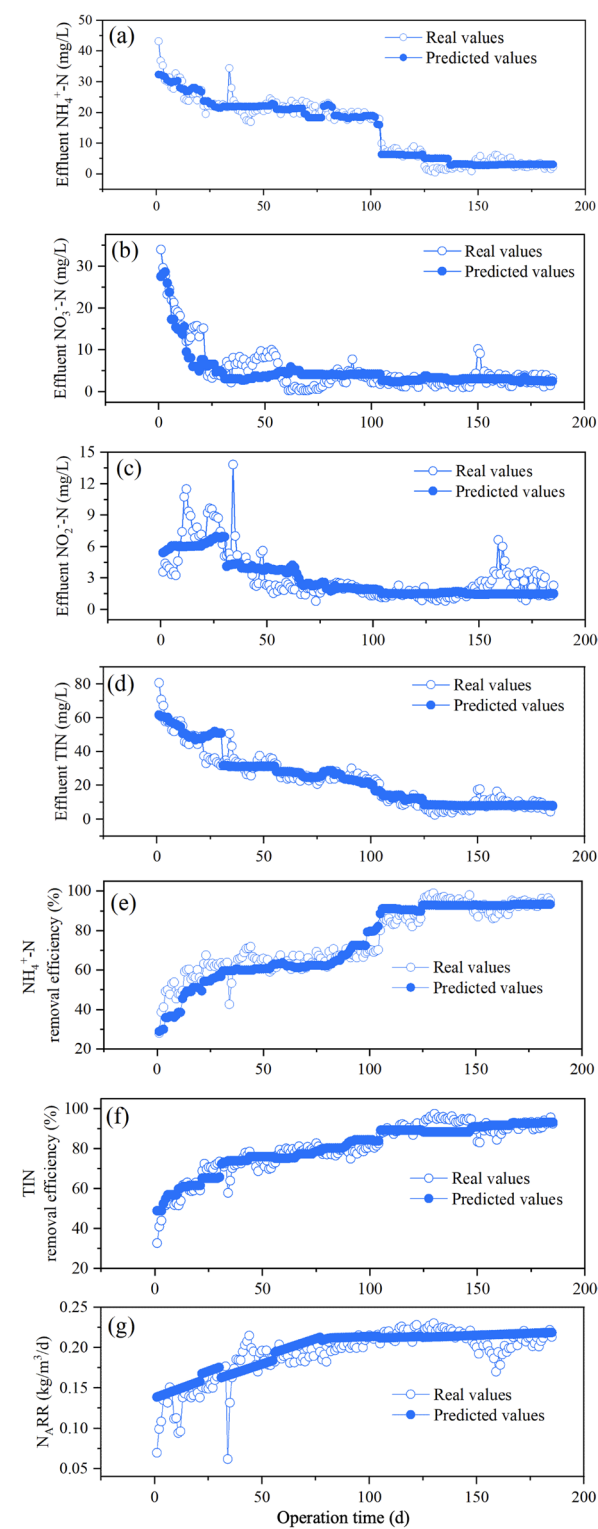


Fig. 5. The predicted values versus real values in the experimental dataset. (a) effluent $\text{NH}_4^+\text{-N}$; (b) effluent $\text{NO}_3^-\text{-N}$; (c) effluent $\text{NO}_2^-\text{-N}$; (d) effluent TIN; (e) $\text{NH}_4^+\text{-N}$ removal efficiency; (f) TIN removal efficiency; (g) N_ARR .

and increase the response of effluent $\text{NO}_3^-\text{-N}$. On the other hand, high COD concentration ($> 316.46 \text{ mg/L}$) will limit the anammox activities⁶⁴, weakening the nitrogen removal efficiency in the anammox-based processes. Meanwhile, the response of effluent $\text{NO}_3^-\text{-N}$ to influent $\text{NO}_3^-\text{-N}$ and influent TIN had a different trend at the beginning of their increase (Fig. S5(d) and (e)). The low influent TIN ($< 55.33 \text{ mg/L}$) led to a high response of effluent $\text{NO}_3^-\text{-N}$ ($6.85\text{--}7.82 \text{ mg/L}$). Then, the response of effluent $\text{NO}_3^-\text{-N}$ gradually increased from 4.41 mg/L

to 7.42 mg/L and from 4.45 mg/L to 5.81 mg/L according to the effects of influent NO_3^- -N and influent TIN, respectively. However, for the response of effluent NO_2^- -N, the influent NH_4^+ -N, influent NO_3^- -N, and influent TIN had similar increasing trends (Fig. S7(b), (c) and (d)). Notably, low NLR ($< 0.95 \text{ kg/m}^3/\text{d}$) resulted in a high response of effluent NO_2^- -N (Fig. S7(e)). With the increase of NLR, the response of effluent NO_2^- -N decreased from 4.57 mg/L to 2.51 mg/L and kept stable.

The 2D PDP further provided the response of effluent NO_3^- -N and NO_2^- -N to mutual effects from pairwise variables (Fig. S6 and S8). The highest response of effluent NO_3^- -N was observed at the beginning operation period with high influent NO_3^- -N (Fig. S6(a)). With the increase in operation time, the anammox-based system gradually enhanced the removal capacity of NO_3^- -N. The NO_3^- -N may be consumed by the nitrate reduction to ammonium pathway, where NO_3^- -N is reduced to NO_2^- -N and NH_4^+ -N⁶⁵. Then, the NO_2^- -N and NH_4^+ -N are subsequently transformed to N_2 via anammox process in these anammox-based systems.

A peak of effluent NO_3^- -N was found at the HRT of 10.47–10.98 h through all operation time (Fig. S6(b)), which was consistent with the 1D PDP of HRT for effluent NO_3^- -N (Fig. S5(b)). Moreover, low HRT ($< 9.95 \text{ h}$) was more suitable for the removal of NO_3^- -N. For influent TIN and influent COD versus operation time, clear valleys of the response of effluent NO_3^- -N were observed with small ranges of influent TIN (59.38–87.76 mg/L) and influent COD (0–137.13 mg/L) (Fig. S6(c) and (d)). Compared with operation time, the mutual effects of HRT with influent NO_3^- -N, influent TIN, and influent COD were more complex (Fig. S6(e), (h), and (i)). Low HRT ($< 9.95 \text{ h}$) was favorable to reduce the accumulation of NO_3^- -N with all ranges of influent NO_3^- -N (Fig. S6(e)), which was agreed with the 2D PDP of HRT versus operation time (Fig. S6(b)). The response of effluent NO_3^- -N was lower at the combined of high HRT ($> 15.64 \text{ h}$) and high influent TIN, as well as the combined of low HRT ($< 9.95 \text{ h}$) and low influent TIN, implying that high HRT may promote the accumulation of NO_3^- -N under low TIN loading (Fig. S6(h)). The influent COD versus HRT had the most complex effects on the response of effluent NO_3^- -N (Fig. S6(i)). Two response valleys were found in the range of influent COD (200.42–305.91 mg/L) and HRT ($< 9.95 \text{ h}$, and 11.50–12.54 h).

Notably, the interactions of different influent nutrients (i.e., influent TIN versus influent NO_3^- -N, influent COD versus influent NO_3^- -N, and influent COD versus influent TIN) had a strong influence on the response of effluent NO_3^- -N (Fig. S6(f), (g), and (j)). Overall, the positive correlation between influent NO_3^- -N concentration and the response of effluent NO_3^- -N was confirmed (Fig. S6(f) and (g)), whereas the response of effluent NO_3^- -N was higher at low influent TIN with all range of influent COD (Fig. S6(j)). This difference may be caused by the high percentage of NH_4^+ -N in the influent TIN. Combining influent TIN of 59.38–87.76 mg/L and influent COD of 200.42–305.91 mg/L reached the lowest response of effluent NO_3^- -N (1.39–1.64 mg/L).

The response of effluent NO_2^- -N had an identical trend in the 2D PDP of operation time versus influent NH_4^+ -N (Fig. S8(a)), operation time versus influent NO_3^- -N (Fig. S8(b)), operation time versus influent TIN (Fig. S8(c)), and operation time versus NLR (Fig. S8(d)). Namely, the long-term operation time extremely reduced the response of effluent NO_2^- -N, suggesting that few NO_2^- -N remained in the effluent of successful anammox-based nitrogen removal processes. The 2D PDP of influent nitrogen and NLR also presented clear trends (Fig. S8(e)–(j)), implying that the effluent NO_2^- -N was strongly related to the nitrogen loading in the anammox systems. PD and PN processes were the main sources of NO_2^- -N in the anammox-based system, whereas the anammox process consumes most of NO_2^- -N⁵⁶. The high influent nitrogen loadings may lead to the accumulation of NO_2^- -N in effluent due to the insufficient activity of the anammox process⁶⁶.

Effluent TIN

For predicting effluent TIN and TIN removal efficiency, the influent TIN and operation time were the most important input variables. Besides, the input variables including influent NO_3^- -N, HRT, influent NH_4^+ -N, and NLR played more important roles in predicting effluent TIN (Fig. 2(d)), whereas the COD-related input variables (i.e., influent COD and C/N), HRT and NLR were more important for predicting TIN removal efficiency (Fig. 2(f)).

In the 1D PDP of operation time, the decreasing response of effluent TIN (Fig. S9(a)) and the increasing response of TIN removal efficiency (Fig. S11(a)) indicated the maturity of anammox-based nitrogen removal processes. Moreover, the short HRT ($< 6.84 \text{ h}$) and long HRT ($> 24.45 \text{ h}$) presented better TIN removal performance (Fig. S9(b) and Fig. S11(e)). For influent NH_4^+ -N, influent NO_3^- -N, and influent TIN, identical response trends of the effluent TIN were achieved (Fig. S9(c), (d) and (e)). The response of the effluent TIN greatly increased at 181.18 mg/L, 50.07 mg/L, and 95.87 mg/L for influent NH_4^+ -N, influent NO_3^- -N, and influent TIN, respectively. However, the increasing NLR did not lead to the growing response of effluent TIN (Fig. S9(f)) because NLR was affected by the mutual interactions of influent TIN and HRT. In the 1D PDP of influent TIN (Fig. S11(b)), influent COD (Fig. S11(c)), C/N (Fig. S11(d)), and NLR (Fig. S11(f)), similar trends of the response of TIN removal efficiency were observed. The highest responses of TIN removal efficiency were achieved at 87.76 mg/L, 179.33 mg/L, 2.95, and $0.49 \text{ kg/m}^3/\text{d}$ for influent TIN, influent COD, C/N, and NLR, respectively.

For the response of effluent TIN, the 2D PDP of operation time versus HRT (Fig. S10(a)) and operation time versus NLR (Fig. S10(e)) had a decreasing trend with the decrease of operation time. Instead, increasing response of effluent TIN with the increase of influent NH_4^+ -N, influent NO_3^- -N, and influent TIN was identified in the 2D PDP of operation time versus influent NH_4^+ -N, influent NO_3^- -N, and influent TIN (Fig. S10(b), (c) and (d)). Moreover, special patterns in the 2D PDP of HRT versus influent NH_4^+ -N, influent NO_3^- -N, and influent TIN were captured, where the low valleys of effluent TIN were achieved at the HRT of $< 8.91 \text{ h}$, 11.50–12.54 h, 15.64–17.20 h, and $> 24.45 \text{ h}$ (Fig. S10(f), (g) and (h)). The 2D PDP of HRT versus NLR further confirmed that high HRT ($> 24.45 \text{ h}$) and low HRT ($< 2.18 \text{ h}$) led to the lowest effluent TIN at high NLR levels (Fig. S10(i)).

For TIN removal efficiency, the 2D PDP of operation time versus influent TIN (Fig. S12(a)), influent COD (Fig. S12(b)), C/N (Fig. S12(c)), HRT (Fig. S12(d)), and NLR (Fig. S12(e)) showed a clearly increasing response

of TIN removal efficiency with the increase of operation time. Meanwhile, low influent TIN, low influent COD, low C/N, and low NLR led to the peaks of TIN removal efficiency. More complex patterns were found in the 2D PDP of influent TIN versus influent COD (Fig. S12(f)), HRT (Fig. S12(g)), and NLR (Fig. S12(h)). The highest response of TIN removal efficiency was observed at the influent TIN = 71.54–87.76 mg/L and influent COD = 200.42–221.52 mg/L (Fig. S12(f)). In the 2D PDP of influent TIN versus HRT, the highest response of TIN removal efficiency was found at the influent TIN = 71.54–87.76 mg/L and HRT = 0.63–6.84 h, as well as the influent TIN > 108.03 mg/L and HRT = 24.45–26 h (Fig. S12(g)). Similarly, the highest response of TIN removal efficiency in the 2D PDP of influent TIN versus NLR was identified at the influent TIN = 71.54–87.76 mg/L (Fig. S12(h)). In the 2D PDP of influent COD versus HRT, the highest TIN removal efficiency was also found at the HRT = 0.63–6.84 h and HRT = 24.45–26 h with influent COD = 168.78–200.42 mg/L (Fig. S12(i)). In the 2D PDP of influent COD versus NLR, the influent COD = 168.78–200.42 mg/L also provided the highest response of TIN removal efficiency (Fig. S12(j)). These optimized influent COD ranges fulfilled the normal COD concentrations in the municipal wastewater^{26,30}, whereas higher TIN concentrations in influent than that in the municipal wastewater were required for the high efficiency of the anammox-based nitrogen processes^{67,68}. These results also provided an ideal range for selecting the parameters (e.g., HRT and NLR) of anammox-based nitrogen removal processes.

N_A RR

N_A RR, as an important calculation variable for anammox-based nitrogen removal rate, was highly affected by the input variables including HRT, NLR, influent TIN, influent NO_2^- -N, influent COD, and influent NH_4^+ -N (Fig. 2(g)). The response of N_A RR had a decreasing trend in the 1D PDP of HRT (Fig. S13(a)) because the HRT was the denominator in the equation of N_A RR (Text S1). In contrast, for nitrogen loading variables (i.e., NLR, influent TIN, influent NO_2^- -N, influent NH_4^+ -N), the response of N_A RR increased (Fig. S13(b), (c), (d), and (f)), suggesting that the contribution of anammox processes was enhanced with higher nitrogen loadings. Meanwhile, except for influent COD > 506.33 mg/L, no obvious variation was observed in the 1D PDP of the influent COD (Fig. S13 (e)). The 2D PDP of influent COD versus NLR (Fig. S14 (a)), influent TIN (Fig. S14 (b)), influent NO_2^- -N (Fig. S14 (c)), and influent NH_4^+ -N (Fig. S14 (d)) further confirmed that a great increase of N_A RR response occurred at the influent COD > 506.33 mg/L. Moreover, an extreme increase in N_A RR response was found at NLR > 1.28 kg/m³/d, influent TIN > 176.89 mg/L, influent NO_2^- -N > 18.78 mg/L and influent NH_4^+ -N > 152.61 mg/L. The combination of influent TIN > 176.94 mg/L and influent NO_2^- -N > 18.78 mg/L led to the highest response of N_A RR (Fig. S14(e)). For HRT versus influent nitrogen and carbon variables (Fig. S14(f), (g), (h), (i) and (j)), an increasing trend of N_A RR response was identified with the decreasing HRT (< 7.36 h). Furthermore, higher N_A RR was found at the high NLR (> 2.18 kg/m³/d) and high influent NH_4^+ -N (> 148.53 kg/m³/d) (Fig. S14(f) and (j)). Overall, the high N_A RR was observed at the high nitrogen loading and low HRT, implying that the anammox-based nitrogen removal processes were capable of adapting to high processing pressure in modern wastewater treatment plants⁶⁹.

Practical implications and limitations

Forecasting the stabilization and operational performance of anammox-based nitrogen removal processes is essential for engineering operations, especially for treating municipal wastewater by anammox-based processes, which were usually unstable in practice^{39,70}. By integrating and comparing the big data collected from different anammox-based studies, this study revealed the difference in data characteristics between experiments using sequencing batch reactor and continuous flow reactor, using floc and aggregate, using synthetic wastewater and municipal wastewater, as well as using different anammox-based process types. Results showed that the anammox-based continuous flow reactors operated by microbial aggregates had great potential to treat high levels of nitrogen pollutants. The big difference between the characteristics of synthetic wastewater and municipal wastewater applied in current experiments extremely weakened the practical meaning of anammox-based experiments based on synthetic wastewater³⁰. Meanwhile, the PNA experiments and PNA combined with PDA experiments showed higher TIN removal efficiency than the PDA experiments. Stronger positive correlations (0.55–0.95) between influent and effluent nitrogen concentrations were found in the PNA experiments, implying the higher sensitivity of PNA processes to the influent characteristics.

Importantly, this study provided a new insight: the fast implementation and high accuracy of the AutoML algorithm for these important output variables (i.e., effluent NH_4^+ -N, effluent NO_3^- -N, effluent NO_2^- -N, effluent TIN, NH_4^+ -N removal efficiency, TIN removal efficiency, and N_A RR) makes it a useful tool to learn the information inside the anammox-based datasets, and predict the practical operation of anammox-based nitrogen removal processes. GBM and XGBoost models were identified as the most accurate machine learning models for predicting the effluent variables in the anammox-based processes (R^2 = 0.814–0.993). The high accuracy of these models was not only achieved for data collected from literature but also confirmed by unseen data from our anammox-based experiment, which proved the excellent generalization ability in predicting the performance of anammox-based nitrogen removal processes.

Moreover, the potential improvement strategies for anammox-based processes could be proposed based on the interpretable results of optimal models generated by the AutoML algorithm. The appropriate ranges of key characteristics such as C/N, influent NH_4^+ -N, influent TIN, influent COD, HRT, and NLR were identified by the 1D and 2D PDP results. To reduce effluent NH_4^+ -N, it is recommended to keep C/N at 2.72–6.32 and influent TIN at 18.85–87.76 mg/L. NO_3^- -N as the by-product of the anammox bacteria accounted for most of the effluent TIN in the anammox-based nitrogen removal processes⁶⁸. Therefore, the conditions for reducing effluent NO_3^- -N in anammox-based nitrogen removal processes deserve deep exploration. Specifically, this study found that the low effluent NO_3^- -N could be achieved at COD of 200.42–305.91 mg/L, HRT of < 9.95 h and 11.50–12.54 h, influent TIN of 59.38–87.76 mg/L, and NLR > 0.95 kg/m³/d. The effluent TIN and TIN removal

efficiency had a similar but more special pattern compared with the effluent NO_3^- -N. The lower effluent TIN and higher TIN removal efficiency were found at the combination of lower influent TIN (71.54–87.76 mg/L) with lower HRT (0.63–6.84 h), and the combination of higher influent TIN (> 108.03 mg/L) with higher HRT (24.45–26 h). It means that a matching HRT and influent TIN concentration should be selected to improve the performance of the anammox-based processes. Meanwhile, a lower influent COD of 168.78–200.42 mg/L was responsible for the high TIN removal efficiency. N_A RR, as an important index for representing the anammox activity, showed a clear increasing trend for higher nitrogen loading and lower HRT. The high N_A RR was obtained at HRT < 7.36 h, NLR > 1.28 kg/m³/d, influent TIN > 176.89 mg/L, influent NO_2^- -N > 18.78 mg/L and influent NH_4^+ -N > 152.61 mg/L. Although this high nitrogen loading was uncommon for municipal wastewater, an appropriate mix of municipal wastewater and industrial wastewater may be helpful to achieve high N_A RR in mainstream anammox-based processes⁷¹. Integrating these data characteristics inside the big dataset could largely deepen the understanding of anammox-based nitrogen removal processes, and effectively reduce the difficulty of managing and operating them.

There are also some limitations existing in this study. First, the prediction accuracy of optimal models generated by the AutoML algorithm suffered from a slight decrease in the prediction of effluent NO_2^- -N, effluent NO_3^- -N, and N_A RR. This decreasing generalization ability may be due to the different operating conditions and microbial compositions in experiments between the training dataset collected from the literature and the unseen dataset collected from our study. Importantly, it was found that effluent NO_2^- -N, effluent NO_3^- -N, and N_A RR, as output variables, were more sensitive to the different operating conditions and microbial compositions in the anammox-based processes. The generalization ability of candidate models generated by the AutoML algorithm may be improved by including more essential input variables related to these output variables in the training datasets⁵⁵.

Secondly, various organics with complex chemical structures exist in municipal wastewater. Among them, some substituents (e.g., $-\text{CH}_3$, $-\text{OH}$) in organics impact the substrate diffusion and membrane stability by changing the hydrophobicity. Moreover, some substituents (e.g., $-\text{Cl}$, $-\text{F}$) in organics contribute to the biotoxicity and will inhibit the anammox activity⁷². However, COD was set as an index for representing all these organics in most anammox-based experiments fed with municipal wastewater. The missing concentrations of these sensitive organics in the datasets will limit the understanding of machine learning models for the effects of different organics on the anammox-based nitrogen removal processes.

On the other hand, although we included the dominant anammox genera in the dataset as a categorical variable, more variables related to microbes should be considered in future studies. Nowadays, the data size of microbes in these experiments is much smaller than the data size of influent and effluent variables. The appropriate ways to collect and utilize microbial data of anammox-based experiments should be further explored to overcome the insufficient data volume⁷³. Employing machine learning algorithms to scrutinize these microbial datasets will not only facilitate a more profound comprehension of the microorganisms involved in anammox-based processes, but also offer valuable insights that enhance the nitrogen removal efficiency of practical anammox-based systems for treating municipal wastewater⁵². Future directions for developing multifactor optimization strategies for anammox treatment should increasingly explore the valuable information hidden in microbial information. Ultimately, the fusion of high-dimensional microbial community fingerprints with process variables will enable the development of biology-informed digital twins, shifting the paradigm of anammox optimization from empirical hydraulic control toward precision microbiome management.

Conclusions

This study explored the response effects of important effluent variables to operation conditions and influent characteristics involved in the anammox-based nitrogen removal processes through the big data analysis and AutoML algorithm. Analysis of big data collected from the literature presented that the continuous flow reactors with anammox aggregates could address higher nitrogen loadings in municipal wastewater. High accuracy of the candidate models generated by the AutoML algorithm for these important output variables was achieved for data collected from both literature and unseen experiments. Moreover, the interpretable analysis of the optimal models clarified the 1D and 2D response effects between input and output variables. The appropriate ranges of key characteristics such as C/N, influent NH_4^+ -N, influent TIN, influent COD, HRT, and NLR were identified. These findings would deepen the understanding and support the practical operation of anammox-based processes for treating municipal wastewater.

Data availability

All data supporting the findings of this study are available within the paper and its Supplementary Information.

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Author contributions

J.L. contributed to algorithm design and improvement, conducted related experiments, and participated in manuscript writing. J.C. provided methodological guidance and contributed to manuscript revision. R.X. conceptualized and supervised the study, designed the experiments, and contributed to manuscript preparation.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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